Predictive Beamforming at 60 GHz

IMM-Anchored minLSTM Tracking with Hybrid DFT-Codebook Beamforming

Predicting where a user will be in the next slot lets the base station aim its beam ahead of time: 75% of the achievable SNR gain (+3 dB for vehicles), measuring 3 beams instead of 128, on single-RF-chain hardware.

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System Model: IMM + minLSTM + 128-Beam Hybrid DFT Beamforming

Noisy Position (0.5 m error) → **IMM Tracker** (3 motion models) → **minLSTM Correction** (learned fix) → **Predicted Position** (next 0.1 s slot) → **Digital-Twin Channel** (DeepMIMO O1_60) → **DFT Beam Selection** (1 RF chain) → **Transmit Beam** (64 antennas)

Built for fast users. The gain comes from **cyclists and vehicles: +0.96 dB and +2.99 dB (80% and 88% of the achievable gain)**. The **IMM** tracks the turns; the **minLSTM** learns only its residual error, with ¼ of an LSTM's parameters; a **128-beam DFT codebook on one RF chain** beats 64 beams (+0.93 vs +0.84 dB) and fully-digital MRT (+0.85 dB). IMM + LSTM is marginally more accurate (0.267 vs 0.279 m) with a tied SNR gain, at about 4× the cost.

1 Motivation

- Large mmWave/THz arrays form narrow beams, about **1.6°** with 64 antennas
- **Reactive** beamforming aims where the user *was*, so the beam goes stale as the user moves
- Re-sweeping all beams costs airtime; **predictive** beamforming aims at the next position instead

2 Experimental Setup

- **600 synthetic trajectories**, 200 slots of 0.1 s; split by whole trajectory (420 / 90 / 90)
- **Pedestrian** (0.8–2 m/s), **cyclist** (3–8 m/s), **vehicular** (8–20 m/s); 0.5 m position noise
- **Fair test:** every beam is scored on the **true** next-slot channel; the oracle knows it (upper bound)

3 Training The Predictor

The IMM Anchor Starts Ahead.

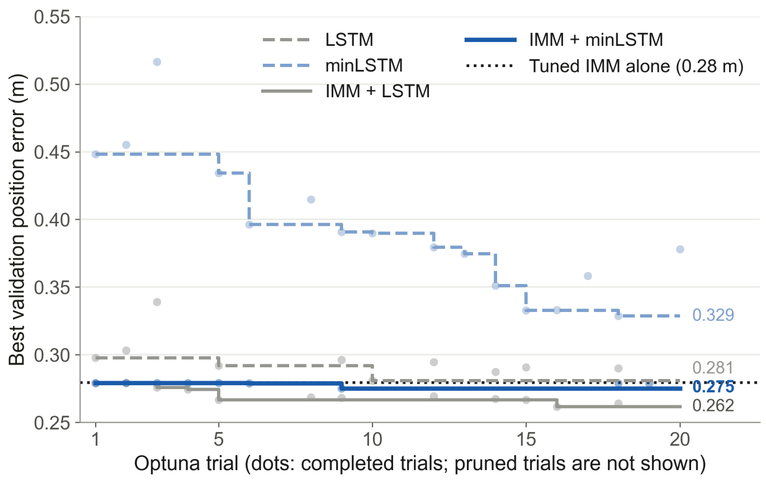


Fig. 1. Best validation error per Optuna trial; same 20-trial budget for every model.

- IMM-anchored models start at the IMM's **0.28 m** on trial 1; they only learn a small correction
- Direct minLSTM must learn motion from scratch: **0.45 → 0.33 m**, still improving at trial 20

Training Is Stable Across Seeds.

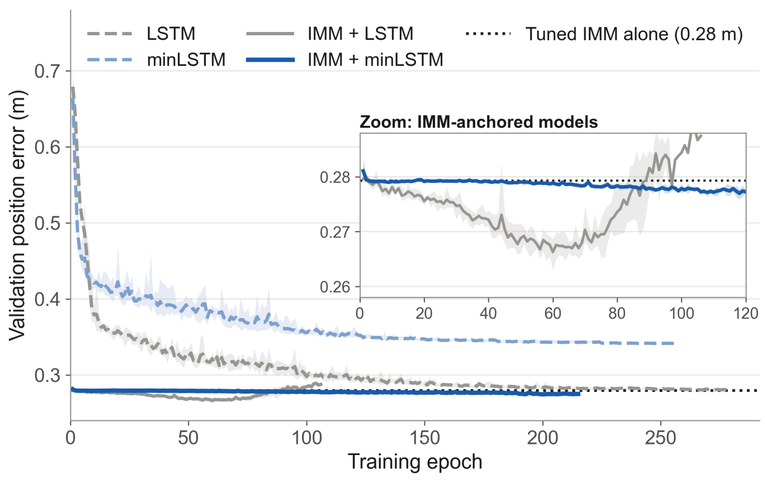


Fig. 2. Validation error per epoch; line = mean of the 5 seeds, band = min–max.

- Direct networks start near **0.68 m** and need 150–190 epochs; minLSTM stops at **0.34 m**
- Anchored models start at the IMM; IMM + LSTM dips to 0.266 m then overfits, **IMM + minLSTM never falls behind the IMM**

4 Prediction Accuracy

One-step predictor	Test error	Params
Hold last position	0.89 m	–
Tuned IMM	0.280 m	–
LSTM (direct)	0.285 m	200k
minLSTM (direct)	0.350 m	100k
IMM + LSTM	0.267 m	201k
IMM + minLSTM (ours)	**0.279 m**	**51.7k**

- Direct networks do not beat a tuned IMM; anchored on it, they do
- **Re-run check:** retrained on a laptop GPU, all four networks reproduce these errors within 2 mm (5 seeds each)

5 Results: SNR And Hardware

Predictive Beats Reactive, Most At High Speed.

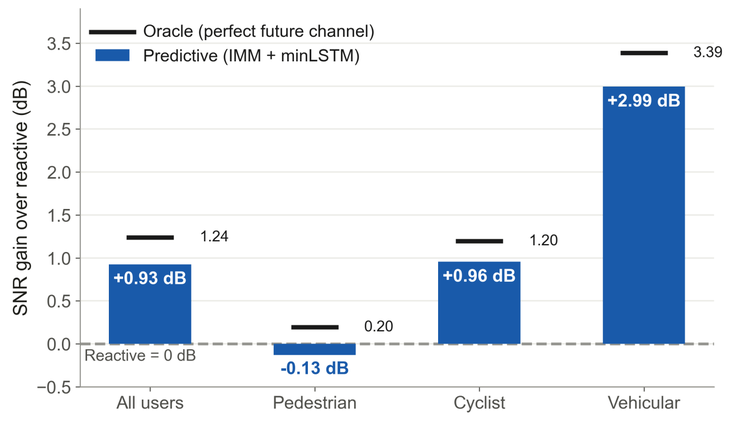


Fig. 3. SNR gain over reactive; hybrid beamforming, 128-beam DFT codebook, 1 RF chain, 64 antennas.

- **+0.93 dB** overall, of 1.24 dB achievable; vehicles **+2.99 dB** (of 3.39), cyclists **+0.96 dB**
- Pedestrians barely move per slot: −0.13 dB (0.20 dB available)
- **Cheap hardware is enough:** one-RF-chain hybrid DFT recovers **75%** of the achievable gain; fully-digital MRT with 64 RF chains recovers 42%

The Gain Survives Hardware Phase Error.

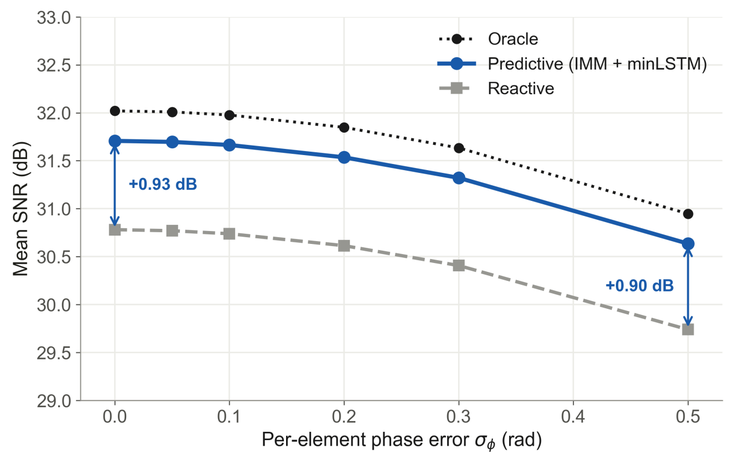


Fig. 4. Mean SNR against i.i.d. per-element phase error, 10 Monte Carlo draws.

- 0.5 rad of phase error costs every method about 1.07 dB; the predictive gain only moves from +0.93 to **+0.90 dB**

6 Robustness

Anchoring On The IMM Keeps Learning Safe.

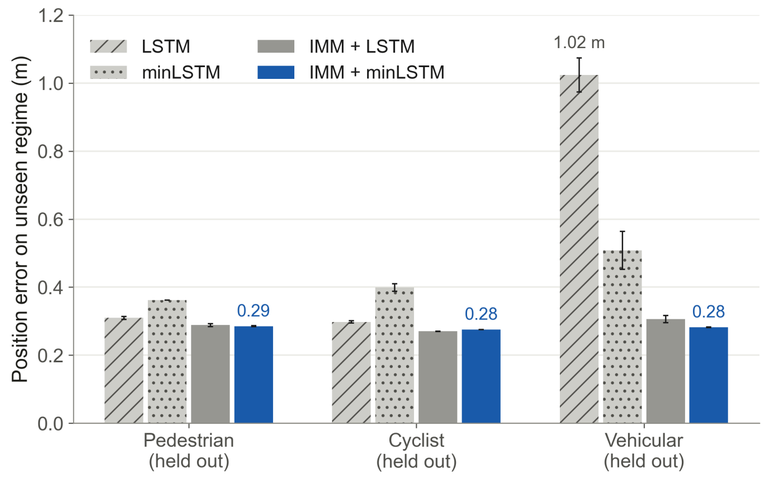


Fig. 5. Position error on a mobility type held out of training (3 seeds).

- On their own, LSTM fails on unseen vehicles (**1.02 m**) and minLSTM reaches 0.51 m
- **IMM + minLSTM stays at 0.28–0.29 m** for every held-out type, the same as the IMM: the anchor removes the failure mode

7 Overhead And Compute

- Misalignment at 128 antennas: **0.51** beamwidths reactive vs **0.17** predictive
- **3 predicted beams reach 31.99 dB**, within 0.03 dB of a full 128-beam sweep
- **Outage** (SNR > 3 dB below oracle): **10.7%** reactive vs **3.0%** predictive
- With beams refreshed every 3.6 ms, a 128-beam sweep leaves no time for data; 3 predicted beams still deliver **10.3 b/s/Hz**

~4×	**¼**	**0.22 ms**
fewer operations per step than LSTM	of the LSTM's parameters	per slot on a CPU, vs a 100 ms slot

Key Takeaways

1. **Safe learning.** An IMM anchor keeps the learned predictor accurate on motion it never saw.
2. **Lightweight AI.** minLSTM has ¼ of an LSTM's parameters and a constant cost per slot.
3. **Right hardware.** One-RF-chain hybrid DFT recovers the most gain on the cheapest hardware.
4. **Less overhead.** Full-sweep SNR from about 40× fewer beam measurements.

Bottom Line

A tuned **IMM does most of the tracking**. A small streaming minLSTM correction on top of it is **safe** (never worse than the IMM, even on unseen motion) and **cheap** (51.7k parameters, 0.22 ms per slot), and pointing a **one-RF-chain DFT beam** at the predicted position recovers 75% of the achievable SNR gain.

Limits

- Synthetic trajectories; one user, narrowband, one slot ahead
- The minLSTM correction adds little over the IMM (0.279 vs 0.280 m)
- No gain for slow pedestrians

Next

- **Real data:** DeepSense 6G and over-the-air SDR tests
- **Longer horizons** with uncertainty-aware beam widening
- **Multi-user, wideband, sub-THz;** an O-RAN xApp

Code and results

DeepMIMO O1_60 ray-traced scenario · minLSTM: Feng et al., "Were RNNs All We Needed?", 2024 · IMM: Blom & Bar-Shalom, 1988

github.com/aarushag02/thz-beamforming